

DSA 3050A — Business Intelligence & Visualization

Key Insights and Recommendations

Eight evidence-backed findings and six operational recommendations

Document	Insights Report
Group	Grp05
Topic	Option 4 — Airline Delay and Operations Dashboard
Dataset	hflights — 227,496 Houston departures, 2011
Source	US DOT Bureau of Transportation Statistics
Members	J. P. Chawanda · Catherine Ingabire · Ruth Musanhu · Faith Chakwanira · Racheal H.

How to read this document

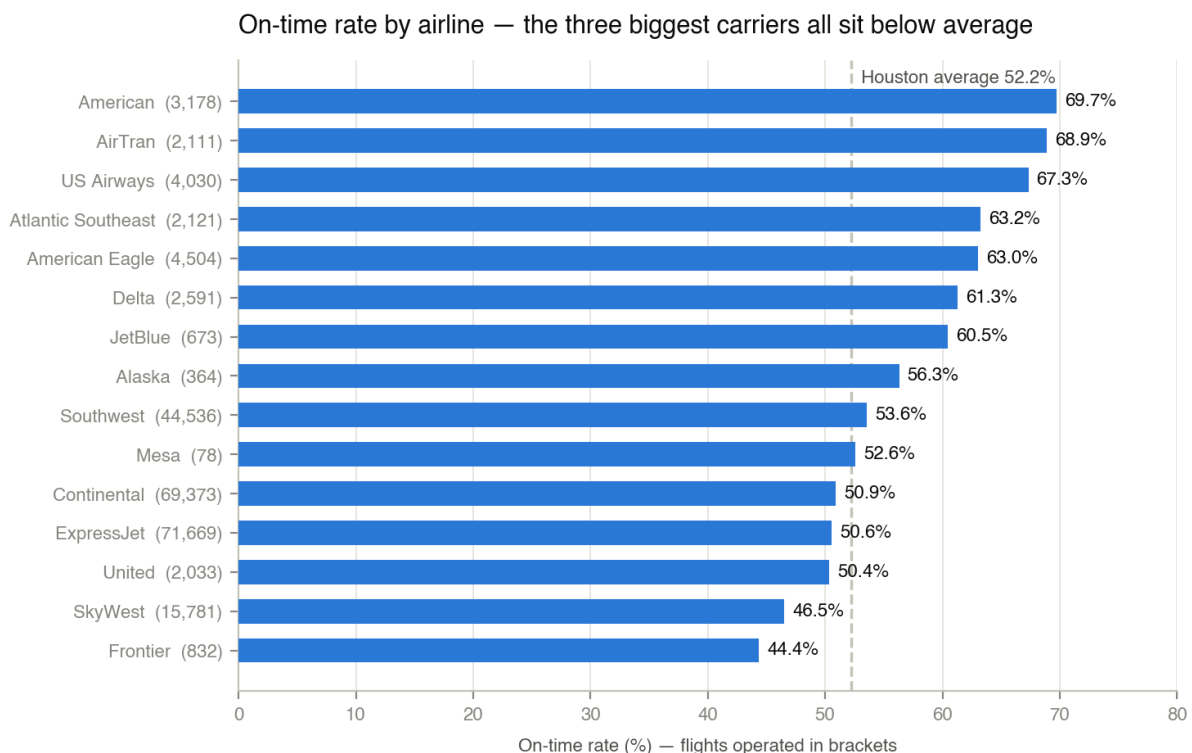
Each insight below states a finding, quantifies it against the 227,496 flights in the dataset, and explains what it means operationally. Every figure is computed directly from the source data and is reproducible by running `Documentation/build_docs.py`. Each insight closes with the dashboard page and visual that evidences it, so any claim can be checked against the report.

Findings are stated as conclusions, not descriptions. "Delay rose in May" is a description; "delay peaks in May while traffic peaks in July, so contingency staffing is aligned to the wrong month" is a finding a manager can act on.

Part A — Key insights

Insight 1 — Punctuality is inversely related to scale — the carriers that fly Houston most are the ones that run it late

ExpressJet (71,669 flights, 50.6% on time), Continental (69,373, 50.9%) and Southwest (44,536, 53.6%) operate 82% of all Houston departures, and all three sit at or below the 52.2% system average. American (69.7%), AirTran (68.9%) and US Airways (67.3%) run more than 15 points better — but on a combined 9,319 flights, which is 4% of the network. The system-wide on-time rate is therefore almost entirely a function of what three carriers do. Improving the other twelve cannot move it. This is not a small-sample artefact: ExpressJet and Continental each fly enough that their rates are stable to within a fraction of a point.

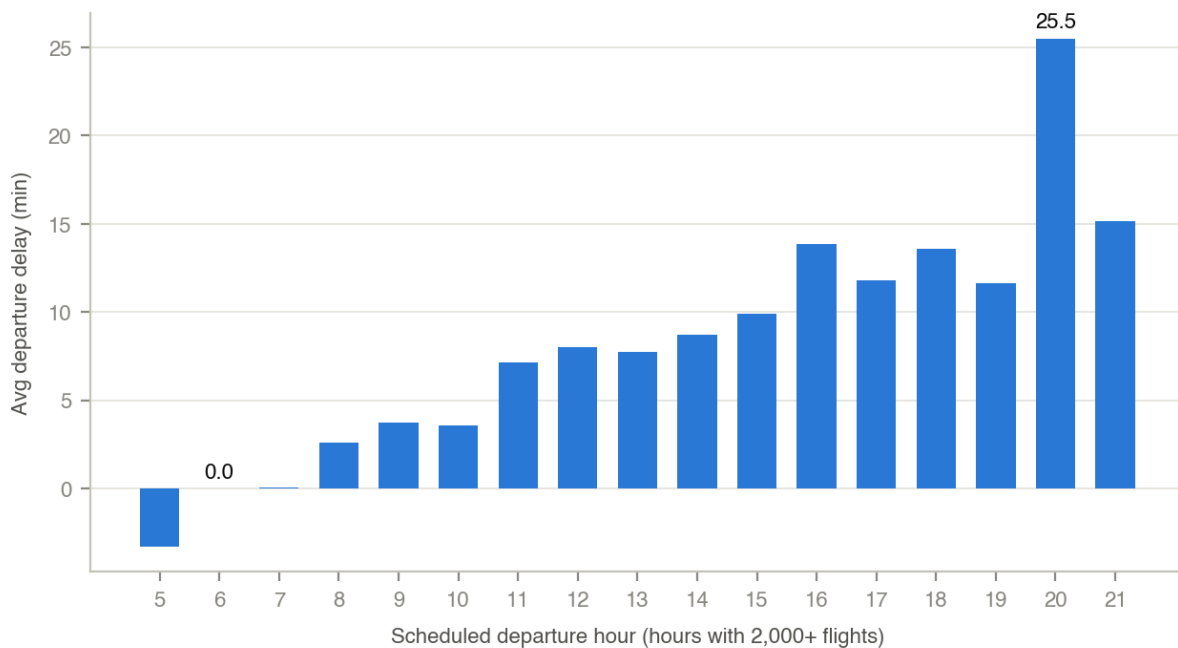


Dashboard evidence: Page 1 — Executive Summary: ribbon chart and airline slicer; Page 3 — airline performance matrix.

Insight 2 — Delay is manufactured during the day, not inherited from the schedule

A flight scheduled to depart at 06:00 leaves, on average, exactly on time (-0.0 minutes). The same network at 20:00 averages 25.5 minutes of departure delay — a 25-minute deterioration with no change in aircraft, crew or route. Delay accumulates monotonically through the day as each late arrival becomes the next late departure. The implication is that late-evening delay is not caused by evening conditions; it is the morning's delay, compounded across four or five rotations of the same airframe.

Delay compounds through the day — a 06:00 departure leaves on time, a 20:00 departure averages 25 minutes late

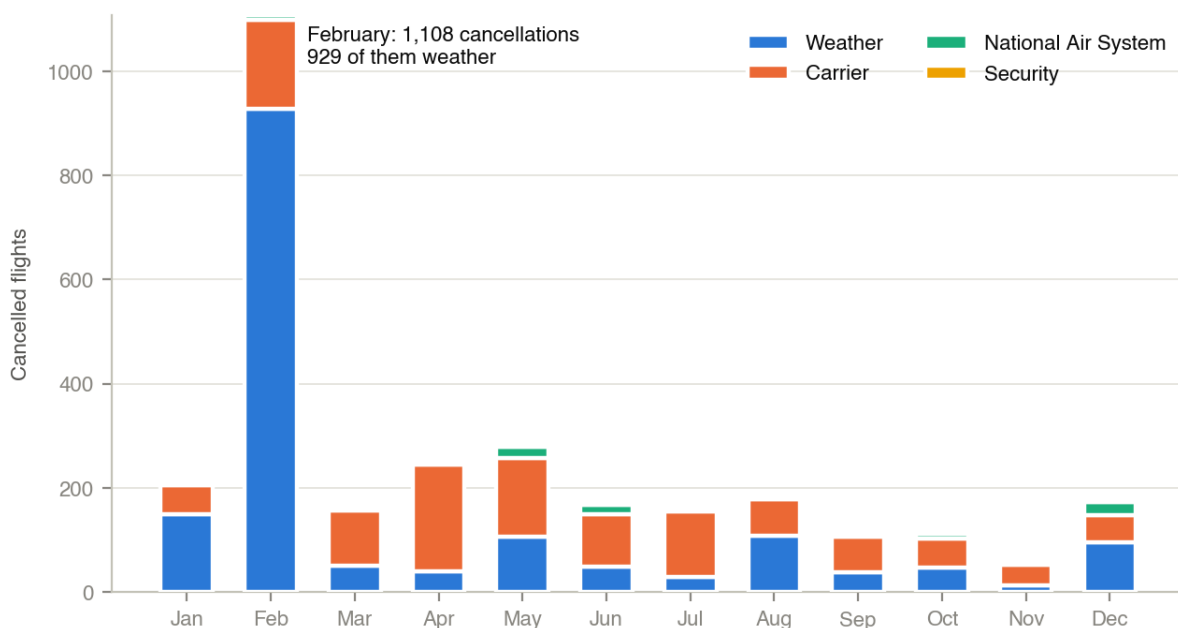


Dashboard evidence: Page 2 — Trend Analysis: time-of-day slicer against the delay trend line.

Insight 3 — Cancellations are not a chronic problem — they are one week in February

Of 2,973 cancellations across the whole year, 1,108 (37.3%) fall in February alone, and 929 of those are coded Weather. February's cancellation rate is 6.47% against a 1.31% annual average — a five-fold spike driven by the Texas ice storm of that month. Strip February out and the network cancels 0.89% of flights, which is a well-run operation. Treating cancellation as an ongoing reliability problem would misdiagnose a single extreme weather event as a systemic failure.

Cancellations are a February problem — one month holds 37% of the year



Dashboard evidence: Page 1 — Executive Summary: decomposition tree expanded on month, then cancellation reason.

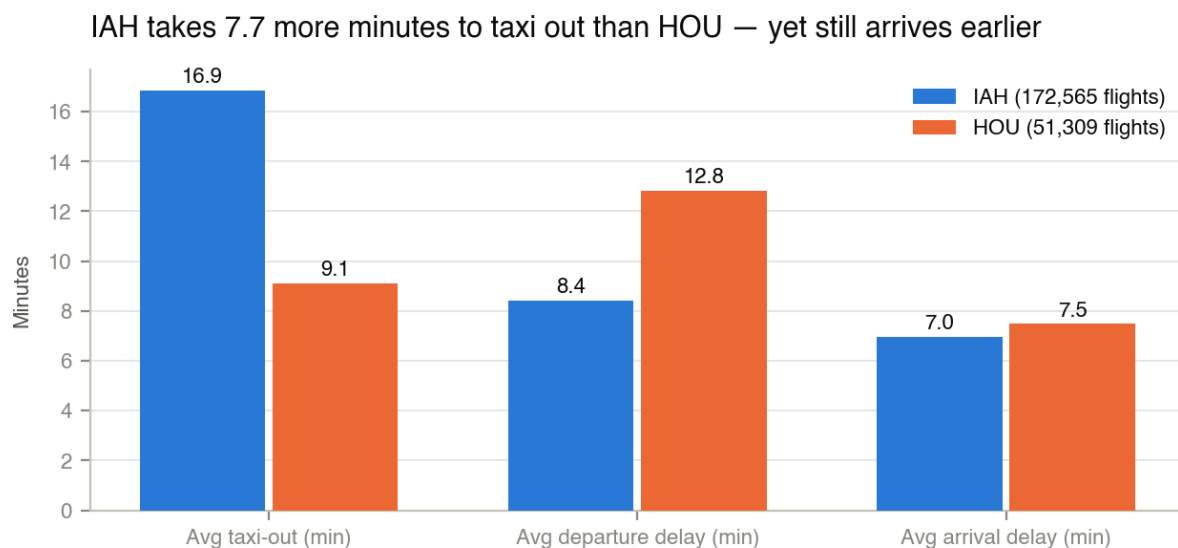
Insight 4 — Carrier-controllable causes dominate every month except the storm

Excluding February, Carrier is the largest cancellation reason in 8 of the remaining 11 months, totalling 1,032 cancellations against 723 for weather. Weather cancellations are concentrated and unavoidable; carrier cancellations are diffuse, persistent and — unlike weather — actionable. The headline reading of the full year (1,652 weather against 1,202 carrier) is an artefact of one month, and it reverses completely once that month is removed. Any conclusion drawn from the annual total alone would point resources at the one cause the operation cannot influence.

Dashboard evidence: Page 1 — Executive Summary: decomposition tree, cancellation reason by month.

Insight 5 — IAH is operationally slower on the ground but still arrives earlier than HOU

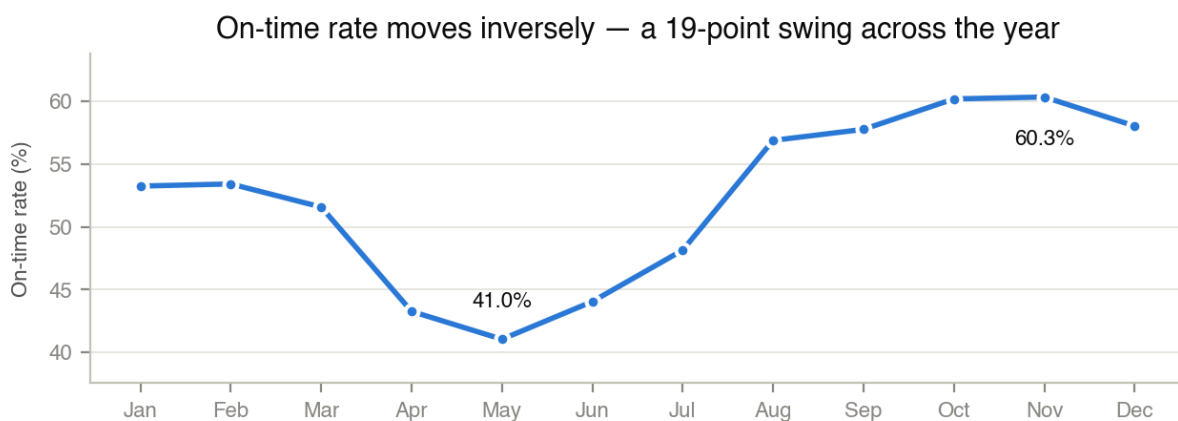
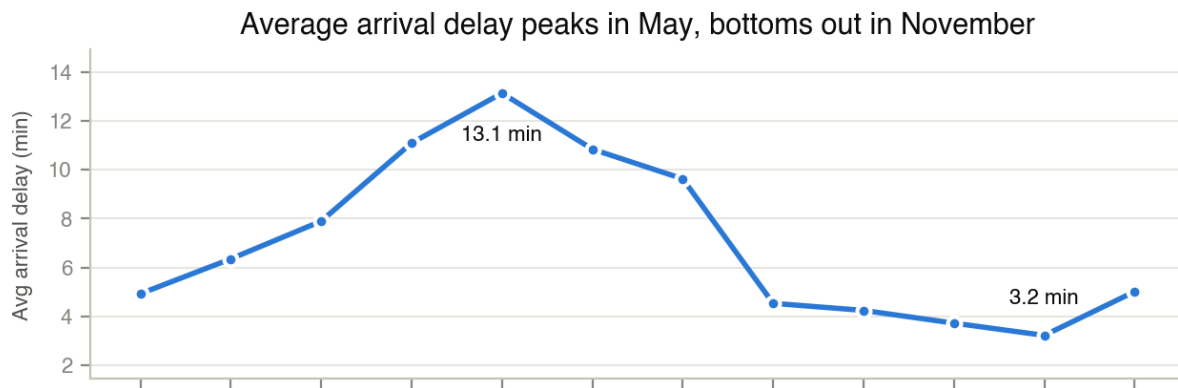
IAH takes 16.9 minutes to taxi out against HOU's 9.1 — nearly double, exactly as expected from a large hub with long taxiways. Yet IAH averages 8.4 minutes of departure delay against HOU's 12.8, and arrives earlier (7.0 minutes against 7.5). HOU's shorter taxi time is being consumed by later pushbacks. The bottleneck at HOU is at the gate, not on the taxiway — which points at turnaround process rather than airfield capacity, and changes what any investment there should be spent on.



Dashboard evidence: Page 3 — Geographic Analysis: origin airport slicer against the KPI cards.

Insight 6 — Delay peaks in late spring, not in summer or at Christmas

Average arrival delay runs from a low of 3.2 minutes in November to a high of 13.1 minutes in May — a four-fold swing — while the on-time rate moves from 60.3% down to 41.0%. July is the busiest month of the year at 20,548 flights yet only the fourth-worst for delay, and December, the month operations plan hardest for, sits in the better half at 5.0 minutes. Traffic volume alone does not predict delay: the correlation between monthly flight count and monthly average delay is $r = 0.33$, accounting for barely a tenth of the variation.

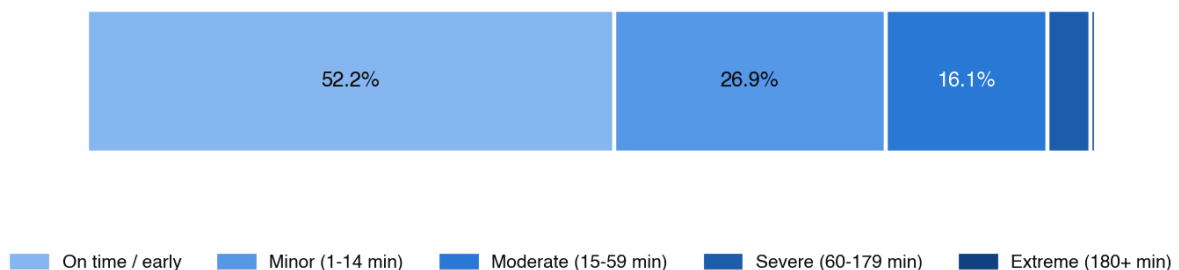


Dashboard evidence: Page 2 — Trend Analysis: monthly delay line with the month slicer.

Insight 7 — Most delay is small, but a thin tail does the damage

52.2% of completed flights arrive on time or early and another 26.9% are under 15 minutes late — 79% of flights are effectively fine. The operational and commercial damage sits in the 4.7% that arrive an hour or more late (10,584 flights), of which 1,110 arrive over three hours late. These are the flights that break connections, breach duty limits and trigger compensation. An average-delay target chases the harmless middle of the distribution; a severe-delay target attacks the costly tail.

Delay severity mix of 223,874 completed flights



Dashboard evidence: Page 1 — Executive Summary: delay category pie chart and KPI cards.

Insight 8 — Two-thirds of departure delay is recovered in the air

Of flights that departed late, 64.9% arrived less late than they departed, and the network recovers 2.3 minutes on average between wheels-up and gate-in. Schedule padding is already absorbing a meaningful share of departure delay. Two things follow: departure delay overstates the passenger-facing problem, and arrival delay is the correct metric for any service-level reporting. It also means there is limited headroom left — the recovery is already being taken, so further departure delay converts almost directly into arrival delay.

Dashboard evidence: Derived from the Delay Recovered column; visible on the drill-through detail table.

Part B — Recommendations

Each recommendation traces to a specific insight and states the expected operational impact.

#	Recommendation	Action	Why it works	From
1	Target the three high-volume carriers, not the fifteen-carrier average	ExpressJet, Continental and Southwest control 82% of departures and all run below average. Establish a joint on-time working group with these three specifically, with a shared target of moving their combined on-time rate from roughly 51% to the 60% already achieved by the mid-size carriers.	Moving those three by nine points moves the entire Houston system by roughly seven. Moving all twelve remaining carriers to perfection would move it by under three. This is the single highest-leverage intervention available.	Insight 1
2	Protect the morning bank and re-time the evening bank	Because delay compounds monotonically from 06:00 to 20:00, the cheapest minute to save is the earliest one. Prioritise first-rotation aircraft for gate, tug and crew resources, and add 10-15 minutes of turnaround buffer to aircraft entering their fourth rotation. Where an aircraft is already 20+ minutes down by midday, swap the tail rather than let the delay propagate.	Attacks delay at the point of creation rather than absorbing it downstream, and stops one late morning aircraft from producing four late evening flights.	Insight 2
3	Fix HOU's turnaround, not its taxiways	HOU has half of IAH's taxi-out time yet 4.4 minutes more departure delay. Do not spend on airfield capacity at HOU. Run a gate-level turnaround study covering boarding process, ground-handling staffing and pushback crew availability.	Redirects capital away from an airfield that is not the constraint and towards the gate process that demonstrably is.	Insight 5
4	Report cancellations with February separated out	Publish cancellation performance as two numbers: a baseline excluding extreme weather events (0.89%) and a weather-event line reported separately.	The blended 1.31% figure hides a well-run operation behind one ice storm, and drives the wrong conclusion — that weather is the dominant cause, when carrier-controllable cancellations are more numerous in 8 of 11 normal months.	Insights 3 and 4
5	Change the operational target from average delay to severe-delay rate	79% of flights are already fine and 4.7% cause nearly all of the cost. Replace the average arrival delay target with a severe delay rate (60 minutes or more) target of under 3.5%, down from the current 4.7%.	Aims resource at the 10,584 flights that break connections instead of at the 1-to-14-minute band, which is largely noise and is already absorbed by schedule padding.	Insight 7
6	Load-plan for May, not for July	Seasonal contingency staffing is currently aligned to traffic volume, which peaks in July. Delay peaks in May, when the network is four times worse than November on a lower flight count. Move reserve crew, standby aircraft and additional ramp staffing into the April-June window and reduce the December over-provision.	Aligns contingency spend with when the network actually fails rather than with when it is busiest — the two are not the same month.	Insight 6

Part C — Storyline

Read together, the eight insights tell one story. Houston in 2011 was not a network with a general punctuality problem; it was a network with four specific, separable problems, three of which are addressable and one of which is not.

The one that is not: February. A single ice storm produced 37% of the year's cancellations. It is weather, it is unforecastable at planning horizons, and no operational change would have prevented it. It should be reported separately and otherwise set aside.

The concentration problem: three carriers fly 82% of the network and all three run below average. Everything the airport does to the other twelve is rounding error.

The compounding problem: delay is created hour by hour through the day, not inherited from a bad schedule. A morning minute saved is worth several evening minutes.

The measurement problem: the operation measures average delay, which is dominated by a harmless 1-to-14-minute band, and blends February into its cancellation reporting. Both choices point management at the wrong target. Changing what is measured is the cheapest of all six recommendations and unlocks the value of the other five.